

ARTIFICIAL INTELLIGENCE FOR ONSITE ANALYSIS OF MICROPLASTICS IN URBAN WATER RUNOFF

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ABSTRACT

This research project addresses the growing problem of microplastic pollution in urban environments by developing an AI state-of-art approach for rapid, onsite analysis. Traditional microplastic identification methods, such as Fourier Transform Infrared spectroscopy, are often time-consuming, and are restricted to in-laboratory analysis. This project aims to develop a cost-effective and portable alternative, combining Nile Red fluorescent staining with artificial intelligence for instant microplastic detection, analysis and quantification. The YOLOv11 object detection model was trained using smartphone captured images of stained artificial plastic samples to identify microplastics based on fluorescence intensity, colour, size, and shape. Some research challenges included the model's ability to detect small objects, differentiate between common everyday plastic types, and handle different fluorescence colours: red, orange, yellow. The study also examined the impact of data augmentation and the possibility of training using artificial samples to detect real urban water microplastic samples. This methodology offers a promising step towards efficient, real-time instant microplastic monitoring in urban water runoff and aid future pollution management.

Some keywords to include:

Nile red, artificial intelligence, object detection, instant recognition of microplastics, YOLOv11, fluorescence.

TABLE OF CONTENTS

ABSTRACT (200 WORDS)	2
INTRODUCTION	4
PROJECT OBJECTIVES	7
EXPERIMENTAL: LAB-BASED	8
OBSERVATIONS	9
EXPERIMENTAL: COMPUTER-BASED	10
RESULTS	11
BACKGROUND THEORY & MECHANISM	15
ACTION OF MECHANISM – NILE RED	15
STRUCTURE OF THE YOLO11	17
SAHI (SLICING AIDED HYPER INFERENCE) ²¹	20
DISCUSSION & ANALYSIS	21
CONCLUSIONS	26
STANDARD OPERATING PROCEDURE (SOP)	27
ACKNOWLEDGEMENTS	30
REFERENCES	31

INTRODUCTION

Urban pollution has become an increasingly widespread problem, especially in recent years following the industrial revolution and associated population growth. Various types of pollution, including air, water, soil, noise, and light pollution, not only harm the environment and human health, but also contribute to a decrease in quality of life. These pollutants pose serious health hazards to human well-being, with plastics and microplastics being among the most harmful.

The use of plastics has been widespread since the "plastic boom" of the 1960s and 1970s, and their polymer breakdown is inevitable. The first signs of plastic pollution in marine life were reported in 1972¹. Carpenter and Smith highlighted the extensive use of artificial plastics in products like facial scrubbers and paint additives as major contributors to microplastics in the environment. The deterioration and mechanical weathering of larger plastic fragments into smaller pieces, causing Pacific gyres and has led to the formation of "The Great Pacific Garbage Patch," a final destination for much unwanted waste plastic². Sources of plastic pollution in the marine environment include illegal dumping, wind-blown household waste, and untreated sewage. Mamun et al.³ detailed the adverse effects of microplastics on human health, covering topics such as plastic pathways, ecological systems, and microplastic contamination entering the human food chain. Marine species such as carp fishes and Atlantic herring have been reported to contain microplastics in their stomachs. Mamun et al.³ also discussed the potential health hazards of microplastics including human infertility, gastrointestinal damage, and carcinogenic effects, concluding that microplastics in the human body pose serious health risks and can cause irreversible damage to human life.

Traditionally, microplastics have been identified through spectroscopic approaches. Techniques used for quantitative and qualitative analysis include Fourier transform infrared (FTIR) spectroscopy, Raman spectroscopy, scanning electron spectroscopy, pyrolysis gas chromatography coupled with mass spectroscopy (Pyr-GC/MS), and liquid chromatography. ATR-FTIR is a common approach, favoured for its characteristic spectral peaks that allow for easy recognition and identification of specific plastic polymer types found in urban water samples. However, limitations include the high cost

and lack of portability of the instrument, restricting analysis to in-laboratory environments. FTIR spectroscopy requires dry microplastic samples, isolation and extraction processes are needed prior to analysis. Raman spectroscopy also share similar limitations. Though coherent anti-stokes Raman scattering (CARS) microscopy can visualize wet microplastics of up to 2-3 μm , this technique remains time-consuming and expensive. Pyr-GC/MS is also primarily limited to laboratory settings due to the need for sample pyrolysis at high temperatures of 700°C. These traditional methods, while highly accurate, are often time-consuming, expensive, and lack portability.

You Only Look Once (YOLO) is the current state-of-the-art, real-time object detection AI model. Unlike traditional methods, YOLO processes object detection tasks as a single regression problem. While early versions had accuracy issues with small objects, the latest versions, YOLOv11, developed by Ultralytics offers open-sourced code, instantaneous object detection speed and accuracy. YOLOv11 incorporated key structural enhancements like a feature pyramid network, anchor-free detection, a path aggregation network, and soft non-maximum suppression. These techniques will allow the prediction of objects of various sizes, which are crucial for the detection of the custom dataset of microplastics.

Traditional methods for microplastic identification such as: Fourier transform infrared spectroscopy, Raman spectroscopy, pyrolysis gas chromatography-mass spectroscopy have been highly accurate but constrained by high costs, laboratory-based techniques and time-consuming processes. These limit onsite analysis of microplastics in urban settings where portability and rapid assessment is essential for pollution management. Fluorescent staining techniques such as Nile Red in combination with artificial intelligence and smartphone photography, allow a cost-effective and time-efficient alternative.

This research project aims to develop an instant recognition of microplastic types based on fluorescence intensity after Nile red staining under the excitation of blue light in an enclosed black box. The system of instant recognition is based on artificial intelligence YOLOv11 (You only look once version 11). By training the YOLOv11 model based on smartphone images of fluorescent artificial samples of various plastic polymer types, the model aims to demonstrate abilities to detect unknown microplastics based on fluorescence intensity, colour, size and shape. Some key challenges and problems of research include: Whether YOLOv11 can spot small object bits captured via a smartphone and its size limitations of the objects. Whether Nile red can accurately highlight between the 7 types of commonly used plastics; PP, LDPE, HDPE, PET, PVC, PS, PMMA and organic material such as grass and cigarette butt filters. Given that if these types of plastics give off different colour fluorescence, whether YOLOv11 can detect the difference between these types of plastic. What the accuracy level and speed of this model is. The limitations of this model and what it can or cannot detect. Given the short time of this project duration, is the model capable of functioning with a limited dataset size with the beneficial use of data augmentation. Whether training the model on artificial macroplastic samples can efficiently be used to detect real urban water microplastic samples.

PROJECT OBJECTIVES

Expanded from the research questions.

- Whether YOLOv11 can spot small object bits captured via a smartphone
 - YOLOv11 can detect large objects with significant object features, solely relying on colour ie. Fluorescence intensity can it also do the same?
 - YOLOv11 is known to be less accurate at detecting small objects, will the additional use of SAHI (Slicing aided hyper inference) be helpful to this model.
- Whether Nile red can accurately highlight between the 7 types of plastic
 - If several plastics give off a similar fluorescence intensity and colour, will the model be able to detect and separate them accurately?
 - Is the colour difference of red, orange and yellow enough to tell the difference between plastic bits?
 - If a different smartphone with different camera settings are used, will the model still be able to detect plastics that fluorescence a slightly different colour?
- What is the accuracy level and speed?
 - How accurate does it detect different polymers? Can it do basic quantification tasks or will small pixelated objects not be counted?
- What are the limitations of this model? What can it not detect?
 - What about false positives such as dust, fibres, undissolved organic matter?

EXPERIMENTAL: LAB-BASED

Experimental methods were followed as per previous research. 10 mg of Nile red was dissolved in 10 mL of ethanol to make a standard solution of concentration 1 mg/mL. Isolated, extracted and purified microplastic sample fragments were stained with 1-2 drops of standard solution in a beaker of 10 mL of deionised water. The artificial plastic samples were cut into and left to stain for approximately 30-60 minutes. Using a filter paper and a Hirsch vacuum glass filtration set, the stained microplastics were isolated and dried. Stained microplastics were transferred to a glass petri dish and a continuous blue light of wavelength 450 nm was used to excite fluorescence in a 3D printed black case. An acrylic orange filter was applied to the black box so that fluorescence wavelengths under 534 nm were blocked, resulting in red, orange, yellow colours only. Fluorescent microplastics were then captured as an image using a smartphone, Samsung A55 5G with a 50MP main camera aided by a 12MP ultrawide camera and a 5MP dedicated macro unit. Various polymer types, organic material and street-found items have been stained with the method outlined above, listed: Polypropylene (PP), polyester (PE), polystyrene (PS), poly terephthalate (PET), low density polyethylene (LDPE), high density polyethylene (HDPE), poly vinyl chloride (PVC), polymethyl methacrylate (PMMA), polyethylene microbeads, cigarette butt filters, plastic nets, grass. FTIR spectroscopy was used to confirm the type of plastics stained, see below Figure 1.

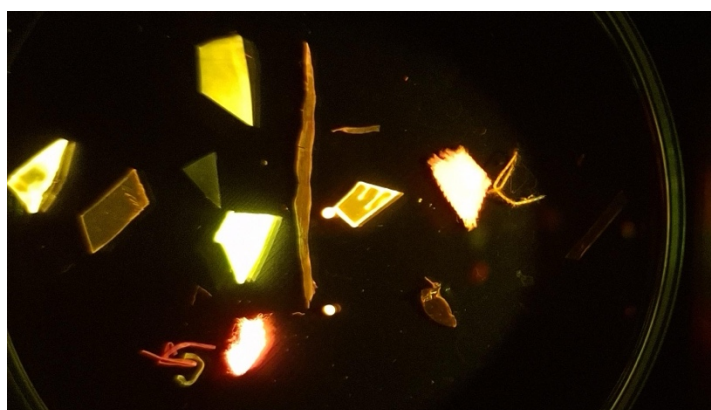
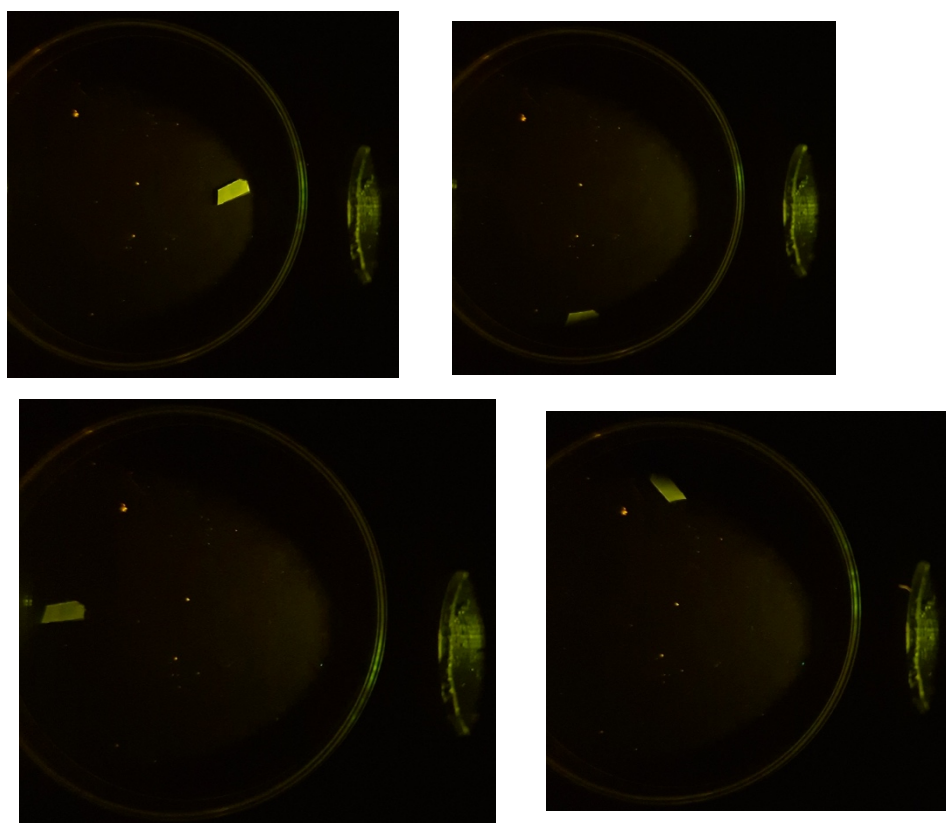


Figure 1. Fluorescence image of artificial sample of plastics. Captured with smartphone camera, iso 250. Ordered from left to right and top to bottom. PS, PET, LDPE, HDPE, HDPE, plastic net, cigarette butt, grass, grass, PMMA, PE microbead, PE microbead, polyester fabric, PVC, polyester fibre, and PP that didn't stain nor fluorescence.

OBSERVATIONS

Originally, under a weak uncharged torch with blue light, the same polymer type but of different colours would give a different colour fluorescence. For example, pink coloured HDPE plastic would give a yellow fluorescence, while black coloured HDPE would not fluorescent. Transparent LDPE would give a light yellow fluorescence while green coloured LDPE would give a light pink fluorescence. Under the excitation of a weak blue light PET plastic would not show fluorescence. However, under the excitation of blue light of a stronger intensity, not only PET plastics would fluorescence but the same polymers of various colours would also stain the same yellow-orange fluorescence. Hence, it is worth noting to keep the intensity of the blue excitation light at the same intensity strength, or, use a somewhat fully charged torch.

Due to the physical placement of the torch in the 3D printed enclosed black box, plastics fluorescent a different tone and cause shadows when placed in a different position on the petri dish. See the figures below, where a piece of PET plastic is positioned right, left, top and bottom:



EXPERIMENTAL: COMPUTER-BASED

Images were labelled using Roboflow to custom train the YOLO11-medium instance segmentation model with 253 layers, 22,368,468 parameters, 22,368,452 gradients, 123.6 GFLOPs. Model weights were trained from random initialisation. Due to the insufficient amount of primary sourced images, the python library 'Albumentations' was used to perform data augmentation. The original library contained 27 high quality images. Using certain methods, new images were generated whilst the labels of the objects were kept the same. Methods used were: affine, random scale, shrink to 5-15% of the original object size, elastic transform, cutout, piecewise affine, grid distortion, gaussian noise, motion blur, hue and saturation value, CLAHE (Contrast Limited Adaptive Histogram Equalization), random brightness and contrast. The Lamda technique from the Albumentations library could not be used due to multiple bug errors in the code. The object class microbeads were left out of the augmentation step due to the nature that microbeads are spherical in shape. Transformation of this object would lead to the object not being spherical. Each transformation mentioned above occurred randomly of various probabilities to add randomness and variety in the newly produced images. 540 images and 540 corresponding label files were generated in 6 hours and 19 minutes.

A yaml file was generated with images set for training, validation and testing. 12 classes were ordered, numbering starting from 0: ['HDPE', 'LDPE', 'PET', 'PMMA', 'PP', 'PS', 'PVC', 'cigarette butt', 'grass', 'microbead', 'plastic net', 'polyester']. Further data augmentation was applied in the yaml file, which enabled mosaic and mix-up augmentation techniques. Various versions of the model were developed using Roboflow each with various amounts of dataset photos. Upon moving YOLOv11 model creation from Roboflow to Google Colab, first runtime system crashed. This was solved by changing the runtime type of the Google Colab notebook from CPU to T4 GPU. SAHI (Slicing Aided Hyper Inference) was used to aid the detection of small objects when testing the model.

RESULTS

Version 4 of the model was tested on unseen real microplastic samples from urban water sewages. As version 4 was trained at 640 pixels, the testing image used was resized to 640x640 pixels and for experimental purposes, the original testing image was also passed through the model to compare the differences in pixels. Predictions and detections were output in the format of a csv file. A higher resolution image i.e. More pixels, led to more detections identified by the model.

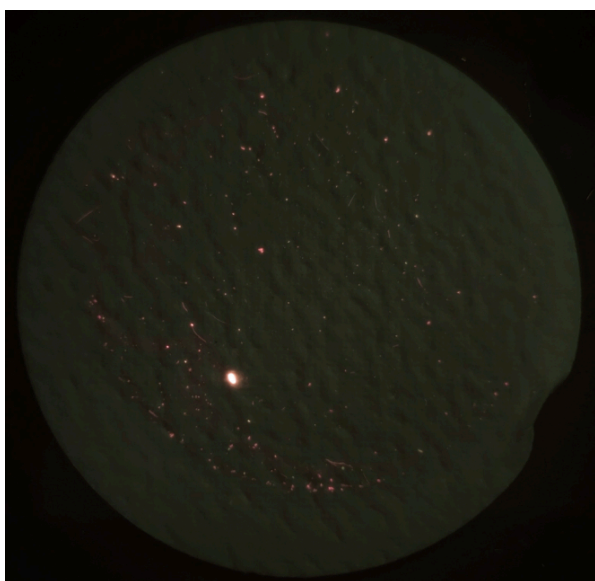


Figure 2. Testing image of microplastic samples from real urban water environments. Image resized to 640x640 pixels. Detection without SAHI. Detected using version 4.

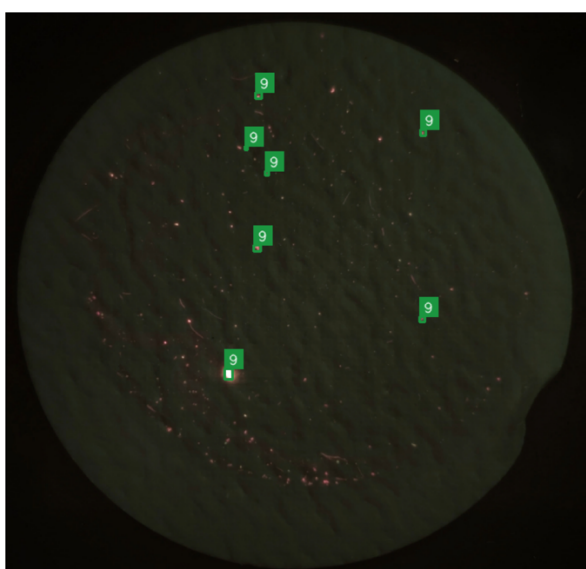


Figure 3. Testing image of microplastic samples from real urban water environments. Image resized to 640x640 pixels. Detection with SAHI. Note, bounding boxes labelled 9 are of object class 'microbead'.

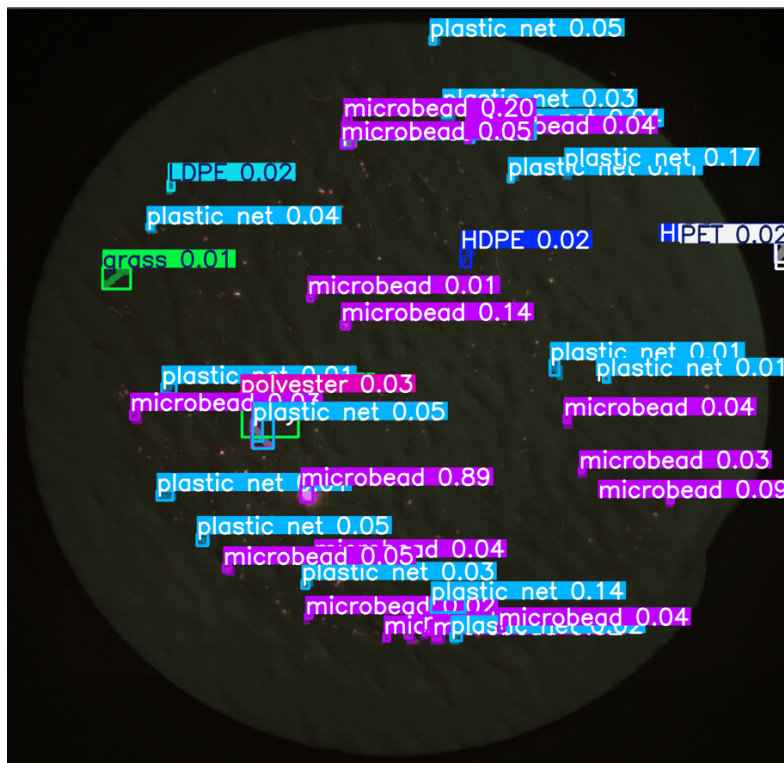


Figure 4. Original testing image of microplastic samples from real urban water environments. Image sized at 3000 x 4000 pixels. Detection without SAHI, bounding boxes labelled with predicted object type and it's confidence level, i.e. Probability of the boxed object being the class object. "Microbead 0.01" would mean 1% probability that the object boxed is a microbead. Model version 5 was used.

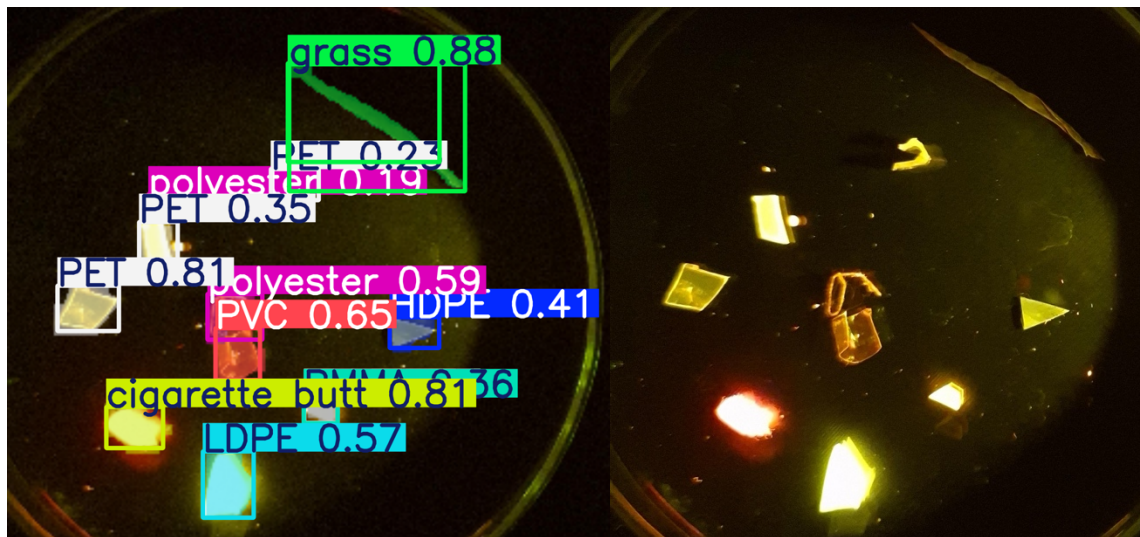


Figure 5. Image of detected and predicted artificial plastic samples passed through the model on the left and image of the original image passed through model. Both images sized at 3000x4000 pixels. Model version 5 was used.

In total 17 versions of the model were created throughout the 12 weeks. The most important versions of the model are listed in Table 1. In general, red fluorescence arise from cigarette butts. Orange fluorescence arise from PET, grass, PVC, and polyester. Yellow fluorescence arise from polyethylene microbeads, PS, HDPE, LDPE, and PMMA. Stained PP did not fluorescence. The project concluded with version 5 as the final version of the model.

Table 1. Table with some of the most important versions of the instance segmentation YOLO11 model created, versions renamed. Note that certain parameters such as epochs, batch size and amount of data augmented photos could not be set on Roboflow.

Versions and platform	Pixel, epochs, batch size	Amount of Original Photos	Amount of Original Photos + Data Augmented photos	Training Time
Version 1 - Roboflow	640, 300, 16	35	92	8 mins
Version 2 - Roboflow	640, 300, 16	149	425	30 mins
Version 3 - Roboflow	2048, 300, 16	405	1173	9 hours
Version 4 - code editor	640, 25, 8	405	929	8.8 hours
Version 5 – code editor	1280, 30, 8	405	1273	11.3 hours

Note that though Roboflow versions can be used online on the Roboflow website, but as the free plan was used to train these models and versions, the weights, graphs, confusion matrices, cross validation test set of these models cannot be accessed. With a large epoch size and a small amount of dataset images, the version 1 and 2 Roboflow models were most likely overfitted leading to unreliable cross validation results.

It was unclear what GPU chips Roboflow used in the server to train versions 1, 2 and 3. Version 4 used RTX 3050 GPU chip. Version 5 used RTX 4070 super GPU chip. Since different GPU chips were used, the training times cannot be linearly correlated photo amount vs training time.

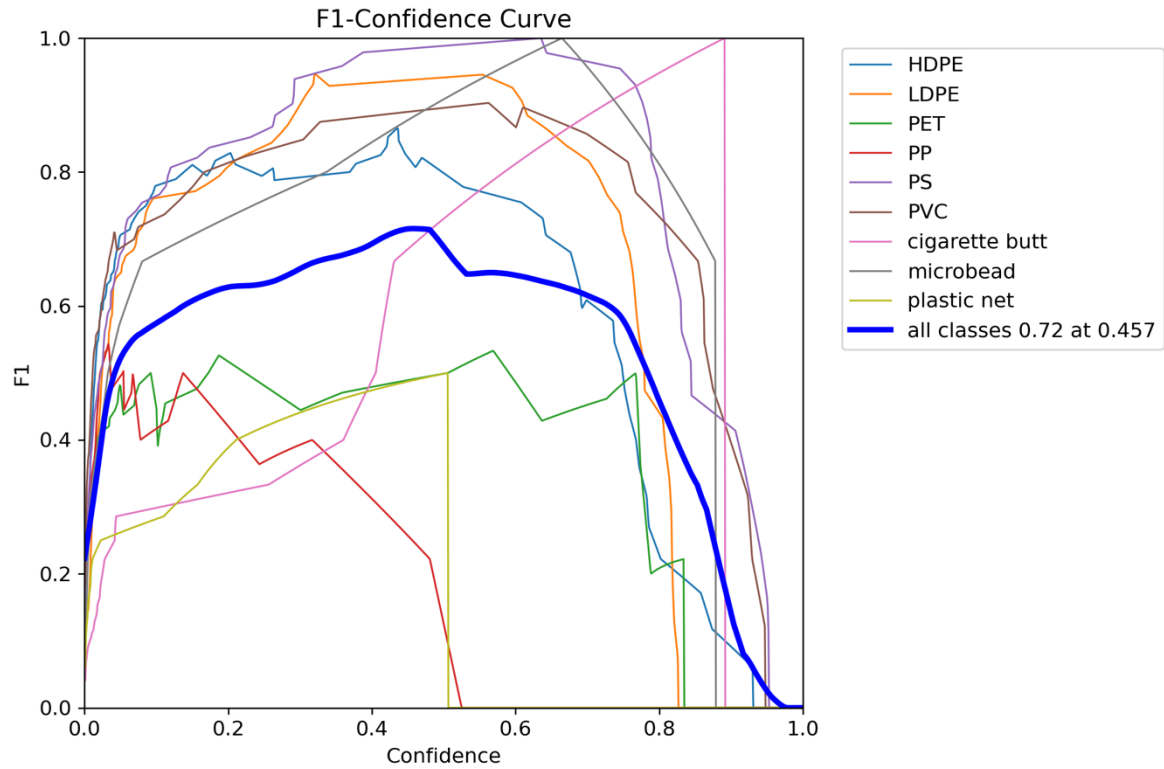


Figure 6. Image of the F1 score confidence curve of the precision and recall of the YOLOv11 model.

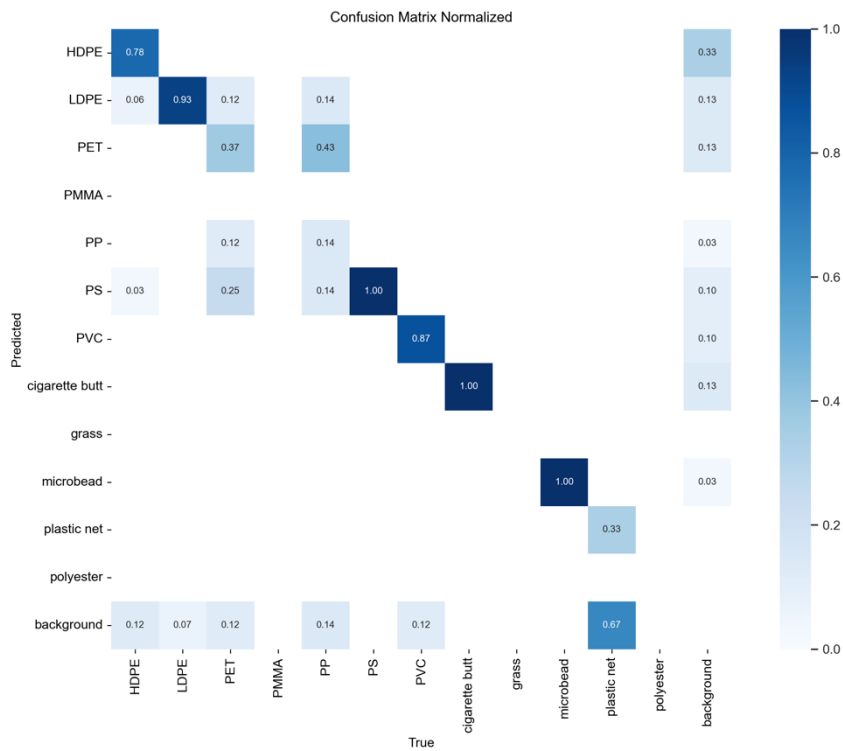


Figure 7. Graph of the normalized confusion matrix. Normalized values of ranges 0 to 1, decimals interpreted as percentages (e.g. 0.78 means 78%).

BACKGROUND THEORY & MECHANISM

ACTION OF MECHANISM – NILE RED

Nile red is a lipophilic dye commonly used to stain lipid cell membranes. A major advantage of Nile red is the inexpensive cost and the high fluorescence properties due to the 9-diethylamino substituent, the electron delocalisation of the structure leads the carbonyl group as the final electron acceptor. In nonpolar environments, such as those provided by lipids or nonpolar organic solvents, Nile Red typically fluoresces strongly in the yellow to gold range, with emission wavelengths generally between 520 and 585 nm when excited with blue light.⁶ As the polarity of the environment and the plastic increases, such as polar plastics in water aqueous environments, the fluorescence emission of Nile Red shifts towards longer wavelengths, resulting in colours near the orange to red spectrum, typically between 585 to 660 nm.⁴ As the fluorescent colour emitted by Nile Red changes towards the orange and red spectrum, the dye becomes less efficient at converting the light it absorbs into the light it emits. This lower efficiency, a decrease in quantum yield, results in the fluorescence appearing dimmer or less bright.⁴ The staining of hydrophobic plastics by Nile Red in an aqueous medium is primarily governed by the principle of hydrophobicity.⁷ Due to its poor solubility in water, Nile Red molecules tend to be pushed from polar aqueous water molecules to the surface of the less polar plastic material.⁵ The hydrophobic effect of Nile Red and favours the bonding of nonpolar molecules in an aqueous environment to minimize the energetically unfavourable interactions between the nonpolar surfaces and water molecules.⁵ The preferential binding of Nile Red to plastics over water is well-established in the context of microplastic detection.⁹ At the plastic surface, Nile Red molecules are physically adsorbed through relatively weak intermolecular forces, primarily van der Waals interactions.⁷ These forces are sufficient to hold the dye molecules onto the polymer surface. Softer, more porous polymers like LDPE might allow for some degree of dye penetration, while the same polymer but more rigid structured, like HDPE might restrict the dye to the surface.¹⁰

The solvatochromic behaviour of Nile Red arises from the dye's electronic structure and its ability to undergo intramolecular charge transfer upon excitation with light.¹¹ The excited state of Nile Red possesses a significantly larger dipole moment compared to its ground state. In a polar environment, the polar solvent molecules can interact with and stabilize this highly polar excited state through favourable dipole-dipole interactions.⁸ This stabilization lowers the energy of the excited state, resulting in a smaller energy difference between the excited and ground states. Consequently, the emitted photons have lower energy and longer wavelengths, leading to a red-shifted fluorescence.⁵ Conversely, in a nonpolar environment, the excited state of Nile Red experiences less stabilization.⁵ This results in a larger energy difference between the excited and ground states, and the emitted photons have higher energy and shorter wavelengths, leading to a blue-shifted fluorescence.¹² In highly polar solvents like water, the fluorescence of Nile Red is significantly quenched due to strong interactions with the solvent and the potential for aggregation.⁴

Thus, theoretically, the non-polar plastics such as LDPE (low density polyethylene), HDPE (high density polyethylene), PE (polyethylene), PP (polypropylene) used in this project should give a yellow/orange fluorescence in ranges of wavelengths 530-600 nm. PS (polystyrene) though non-polar, contains an aromatic ring which allows pi interactions should fluorescence somewhat near the orange colour range. PET (polyethylene terephthalate) and polyester are polar and should give an orange to red fluorescence, somewhat near the range of colours orange to red of around 585-660 nm. PVC (polyvinyl chloride) is highly polar due to the carbon-chloride bonds, should give a red fluorescence of wavelengths roughly 600-660 nm.

STRUCTURE OF THE YOLO11

YOLOv11 (You Only Look Once version 11) is the current state-of-art, high-performance, single-stage object detection model.¹³ The structure of the YOLOv11 training and detection systems begins with an input image, a typical smartphone captured image would consist of a 3000x4000 pixel RGB image, which is then resized to 640x640 pixels or 1280x1280 pixels, to reduce computational cost during the training of the YOLOv11 model. Each input image is composed of 3-channel layers: Red, Green, Blue, where each pixel in each layer consists of pixel values ranging from 0 (black) to 255 (white). Values between 0 to 255 in the red channel layer would represent a shade of red colour. The resized image is then passed into the backbone of the model, with aims to produce a feature map from the input image, performing feature extraction. YOLOv11, like previous Ultralytics models and other object detection models, relies heavily on convolutional neural networks (CNNs) for feature extraction.¹⁴ These networks scan the image using learnable filters called kernels, a matrix of a certain size i.e. 3x3 grid or 10x10 grid, which extract low-level features such as edges, gradients, and textures in the early layers, and higher-level semantic features in deeper layers. The core operation of the kernel (or 3x3 matrix) with the input image. Involves the process of sliding the kernel across the image, performing element-wise multiplication, and summing the results to produce a feature map Mathematically, for a 2D image I and a kernel K , the convolution operation is defined as:

$$S_{(i,j)} = (I \cdot K)_{(i,j)} = \sum_m \sum_n I_{(i+m,j+n)} \cdot K_{(m,n)}$$

Where: $S(i, j)$ is the output feature map. I is the input image. K is the kernel. i, j are the coordinates in the output feature map. m, n are the coordinates in the kernel.

YOLOv11 uses the SPPF (Spatial Pyramid Pooling – Fast) technique, which uses pooling operations with different kernel sizes to aggregate spatial information from various receptors.¹⁵ This enables the model to detect both small objects extracted by small kernels and large objects extracted by large kernels. YOLOv11 uses C3k2 blocks and C2fPSA (Cross Stage Partial with Parametric Self Attention) in its backbone.¹³ These module were developed by Ultralytics with main features to improve gradient flow, use of 3x3 kernel and reduces computational redundancy, thus allowing the network to

focus on key regions through attention mechanisms, enhancing feature representation. Ultralytics did not publish any literature of the in-depth specifics of the C3k2 and C2fPSA blocks. After passing through the CNN and pooling layers, feature maps of different resolutions, were produced. These feature maps of various sizes are then passed into the neck of YOLOv11. The neck performs feature aggregation which merge information from high-resolution feature maps that capture small details, with low-resolution maps that preserve large outlines and broader context. YOLOv11 uses techniques like FPN (Feature Pyramid Network)¹⁶ and PAN (Path Aggregation Network) ¹⁷, and applies upsampling to increase the spatial resolution of certain feature maps. FPNs are designed to create a multi-scale feature pyramid, enabling the network to detect objects at different scales. It addresses the challenge of detecting small objects, which are often poorly represented in deep feature maps. FPNs create a top-down pathway, where high-level semantic features are propagated down to lower-level feature maps. Lateral connections are used to merge feature maps from the top-down pathway with corresponding feature maps from the backbone. Path Aggregation Network (PAN): PANs enhance FPNs by adding a bottom-up pathway, which propagates low-level positional information to higher-level feature maps. This helps the network to more accurately localize objects. PANs introduce a bottom-up pathway that aggregates features from lower levels to higher levels, in addition to the top-down pathway of FPN. This allows information to flow both ways, improving feature representation. Feature maps produced by FPN and PAN techniques are then fused through matrix concatenation operations, which enable a more refined feature map to be produced.

The head of the model uses this refined feature map and use the C3k2 and CBS (Convolution-BatchNorm-SiLU) blocks. The head is divided into detection head branches, each are responsible for predicting object bounding boxes, class probabilities, and confidence scores. YOLOv11 uses an anchor-free detection mechanism ¹⁸, meaning it doesn't rely on preset bounding box shapes. Instead, it learns to predict bounding boxes directly from spatial features in the image from the feature maps. This results in many potential bounding boxes. Non-Maximum Suppression (NMS) is then used to remove most of the generated bounding boxes, leaving the most probable box in the final output image. ¹⁹ NMS compares how much the boxes overlap using the

Intersection over Union metric and keeps the boxes with the highest confidence scores. In doing so, the model does not need to 'save the location' of possible objects, instead, it directly generates a bounding box and removes them in later stages, saving computational effort. During detection, the final output image consists of: bounding boxes, the predicted class, and the confidence level of the predicted class is true in the bounding box.

During the training process of YOLOv11, the model compares its final predictions of a training image with the ground truth of that labelled image, i.e. comparing the bounding box and class of the object with the bounding boxes and the classes of the same labelled image. The comparison is done via loss functions, Stochastic Gradient Descent and Adaptive Moment Estimation.²⁰ The loss functions are essentially differential equations with aims to achieve the lowest non zero value between the predicted object box and the actual object box. The lower the loss value, the closer, more accurate the predicted object is with the actual object. Bounding box loss (the position of the box in correspondence to the object), class loss (the type of object predicted) and segmentation loss (the outline of the actual object) are calculated by the computer, which are then used to readjust the weights of the neural network layers of YOLOv11.

During the detection or inference phase of the YOLOv11, the model operates in a feedforward-only manner, which increases the speed of detection as there is no backpropagation process because the weights are set and there is no need to readjust the weights. During the training stage the model use loss functions and gradients to adjust its weights through backpropagation, a time-consuming process. Each training cycle (epoch) of the data involves calculating the loss function and adjusting the weight to achieve the lowest non-zero value, and thus improving the model's predictive accuracy over time.

SAHI (SLICING AIDED HYPER INFERENCE) ²¹

SAHI is not a model, it is a technique developed to tackle the challenge of small object detection. This is because YOLOv11 models undergo significant downscaling when processed by standard object detection models that typically require a fixed input size. This downscaling can lead to the loss of critical visual features necessary for accurate detection. The SAHI method revolves around a defined function that uses slicing in the detection stage of the model. The input image is divided into overlapping patches which significantly increase the pixel area available for small objects compared to the original images processed by the model. Afterwards, these predictions are converted back into the original image using Non-Maximum Suppression to reduce the number of overlapping boxes and the final amount of bounding boxes and its predictions are represented in the output image.

DISCUSSION & ANALYSIS

As discussed above, non-polar plastics such as LDPE, HDPE, PE, PP theoretically should give off a yellow-orange fluorescence. PS should fluorescence an orange colour. PET and polyester should give an orange-red fluorescence. PVC should give off a red fluorescence. However, in experimental practice, LDPE, HDPE, PE, PS, PMMA gave a yellow fluorescence. PP though non-polar stained with Nile Red, did not fluorescence. PET gave a yellow-orange fluorescence, PVC gave an orange fluorescence, and cigarette butts gave a red fluorescence as shown in Figure 1 above. The difference in wavelength fluorescence could be due to the camera settings used to take the fluorescent microplastics. A higher white balance was set on the smartphone camera used in the experiments, hence all the microplastics gave a somewhat lighter fluorescent tone. However, this experimental and theory indifference could also be due to the various colour coatings of the plastics used in the experiment.

The final version of the YOLOv11 model, version 5, trained with parameter settings at 1280×1280 pixels at 30 epochs in batch size of 8, demonstrated a rather well performance at both artificial sampling and real microplastic samples object detection, as shown in Figure 6 and 7. The F1 confidence curve in Figure 6, shows the F1 score which is a representative value of the precision and recall of the model.

$$F1\ score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Where Precision is the proportion of correct predictions made by the model, i.e. $Precision = \frac{True\ positives}{True\ positives + False\ positives}$ and Recall is the proportion of correctly identified ground truth objects i.e. $Recall = \frac{True\ positives}{True\ positives + False\ negatives}$.

In this way, the overall performance of the model can be analysed. As shown in Figure 6, HDPE, LDPE, PVC, PS and microbead classes perform similar to a normal distribution, where there is high F1 scores i.e. high precision and recall for the bounding boxes at confidence levels between 0.1 to 0.8. There are irregularly performing classes such as PET, cigarette butt filters, PP, plastic nets, where precision and accuracy are generally low regardless of confidence levels. This can be interpreted as not enough quality dataset images that contain these classes, which leads to an underperforming object

detection model when used to predict these classes. A dataset of higher quality and variability with these classes can perhaps improve the F1 precision, accuracy and recall score of these specific classes. However, in general, the thick blue line indicated that all classes performed at an F1-score of 0.72, predicting objects at 0.457 confidence levels, i.e. the model is 72% accurate when giving predictions of confidence at 45.7%. This indicates plenty room for model improvement using higher quality datasets mimicking both real urban water runoff microplastic samples and a wider variety of artificial plastic samples.

Furthermore, the normalized confusion matrix shown in Figure 7, shows the performance of the custom trained YOLOv11 model to correctly identify the class of an object when compared to the ground truth of the labelled trained image. In Figure 7, the diagonal represented the normalized value ranging between 0 to 1 of the true class compared to the predicted class, i.e. On 78% of the occasions, HDPE was correctly predicted. The diagonal in Figure 7 shows room for improvement with classes such as PET, PP, plastic nets, PMMA, grass, polyester and the background. This could be due to PET, PP and plastic nets not giving off a strong fluorescence stain in the photos used to train the AI model in this project experiment. The lack of normalized values for PMMA, grass, polyester and background predictions could be due an error in the maths calculation when normalizing the values, or could be due to the small amount of labelled datasets present in this project. The count for the object classes were as follows:

Table 2. Table with the object count of each classes as listed on the labelling platform, Roboflow.

Object Class	Object Count	Object Class	Object Count
HDPE	352	Cigarette butt filters	246
LDPE	286	Grass	117
PET	280	Microbead	1005
PMMA	100	Plastic net	128
PP	132	Polyester	114
PS	546	PVC	283

Hence, it could be assumed that perhaps due to small amount of object counts for the classes: PP, plastic nets, PMMA, grass, and polyester is the reason for the absence of normalized values or a low normalized values in the confusion matrix. Furthermore, only objects are labelled in images, which means that the background of each image are not explicitly labelled. Perhaps, this is the reason for the absence of a normalized value in the confusion matrix in Figure 7. It is worth noting that classes such as PS, cigarette butt filters and microbeads performed exceptionally well. This could be due to cigarette butt filters being the only source of red fluorescence present in the small list of object classes and the distinct spherical shape of the microbeads class amongst the small list of object classes. With a wider range of object classes and possibly more objects giving off a red fluorescence colour, could lead to the decrease in this accuracy of the cigarette butt filter object class detection. Also, PS and microbead class object count of 546 and 1005 respectively could be the explanation as to the exceptionally well performing object detection in these classes. This means, perhaps, if all object classes have a higher yield in object counts, with an increase in images dataset, the object detection accuracy for these classes will be higher.

Initially, Roboflow was used as a quick tool to check whether a small set of images along with data augmentation techniques were enough to create a somewhat reliable and functioning model. As mentioned in the Results section above, the two most prominent versions of the custom trained AI model were the: 640×640 version, version 4, and 1280×1280 version, version 5, where the code and images created from scratch in Google Collaboratory. Each version was tested both with and without the use of SAHI (Slicing Aided Hyper Inference). Figure 2 is the testing image of a real urban water microplastic sample using the 640×640 pixel version without SAHI. As seen in Figure 2, no predictions were made, meaning that the 640×640 pixel YOLOv11 version failed to detect any objects in the testing image. Figure 3 is the testing image of the same real urban water microplastic sample using the 640×640 pixel version with SAHI. As seen in Figure 3, 7 predictions that the objects detected were of the microbeads class were made. In this way, it could be seen that SAHI helped with the small object detection in the lower resolution version of the YOLOv11 model. Comparing Figure 2, the low resolution, 640×640 pixel version, with Figure 4, a higher resolution, the 1280×1280 pixel

version. Without the use of SAHI, the higher resolution version of the YOLOv11 model managed to successfully make more object detection predictions, though not very confident. Note that during the computational experimental process, there were issues in implementing the SAHI method via library installation and so SAHI was implemented by defining an actual function in the python script instead of pip installing the SAHI pillow library. SAHI failed to make further improvements in predictions with the 1280×1280 pixel version. As previously mentioned in the background theory section, during the custom training stage of YOLOv11, images are resized to either 640×640 or 1280×1280 pixels before inputting them into the CNN of the YOLOv11 backbone, regardless of their original image resolution. However, the comparison between Figure 2, 3 and 4 explained the point that by training the YOLOv11 model with a higher resolution set for example at 3000×3000 pixels, there may not be the need to use SAHI and the accuracy of the detections will be higher. Unfortunately due to computational time and cost of a capable GPU chip, this may not be feasible. Version 3 of the model was trained in Roboflow at 2048×2048 pixel resolution and would be very interesting to compare the models at different resolutions. However, since the free plan was used in Roboflow, the weights, graphs, confusion matrices, and cross validation test sets were inaccessible. Assumingly, Roboflow models would've performed better because an epoch of 300 was set, but Roboflow models would've had a risk of overfitting to the artificial plastic samples and would've performed less accurate to unseen real urban water microplastic samples.

Figure 5 can lightly be used as a cross validation to the model, in this figure, PET 0.35, Polyester 0.19, the two bounding at the grass object, the undetected microbead next to PET 0.35, and PET 0.23, were falsely represented. Once again confirming the values of the normalized confusion matrix of the classes LDPE, HDPE, PVC, cigarette butts, were well trained. Classes such as PET, Polyester, PMMA, need more training with the model.

Figure 5 reinstates that at a large enough size, the detections and predictions made by YOLOv11 are higher than those of a smaller size. Furthermore, since both the 640×640 and the 1280×1280 versions of the model were trained on artificial plastic samples, using these models to detect real urban water microplastic samples are quite a far stretch. This phenomenon is also known as “Garbage in, Garbage out” or “Train with what you

aim to detect”. The foundation of this, lies on the fact that feature maps generated during the training process in the backbone and the neck of the YOLOv11 structure, need to be accurately representing the objects the user wishes to detect. If the aim is to detect real urban water microplastic samples, then it is highly desirable that known real urban water microplastic samples are used to train the model.

With reference to the project objectives mentioned earlier in the report, YOLOv11 is indeed capable of detecting small objects captured by a smartphone, if these objects were represented accurately in the feature maps generated during the training stages. The models trained in this project, version 4 and 5, have been able to identify plastic samples based solely on fluorescence colour of just red, orange and yellow. The use of Nile red have been able to bind differently to 7 different types of plastics and their fluorescence has been distinguishable by the YOLOv11 model. The use of SAHI has indeed been helpful in detecting smaller objects, but only to a certain extent. The feature maps generated during the training stage remain the key to successful small object detection. The use of data augmentation and training with distorted images has made the model robust enough to detect fluorescence plastic objects captured by other smartphones. The nature of the YOLOv11 model has allowed both detection and quantification tasks to occur simultaneously. The custom trained YOLOv11 models were trained on artificial plastic samples so it has not seen potential false positive objects such as dust, fibres, undissolved organic matter as it would potentially see in isolated urban water microplastic samples. However, the trained version 4 and 5 can identify certain organic matters such as grass and cigarette butt filters, as they were object classes trained during the training stages.

CONCLUSIONS

All in all as discussed above, limitations to the model included the low confidence levels in detecting real urban water microplastic samples, and the inaccuracy in detecting the object classes: PP, PET, PMMA, grass, plastic net, and polyester. These limitations could potentially be overcome by training the model on real urban water microplastic samples, so the model is training in what it aims to detect. The use of a supercomputer from the Irish Centre for High-End Computing (ICHEC) could potentially train the YOLOv11 model without reducing the pixel resolution of the training images. In this way, dust-like microplastic particles can be 'seen', detected and analysed by the model. An increase in the set of data augmented images would also make the model more robust, especially when detecting photos captured by other smartphones.

The results concluded from this final year project were reasonably well given the timeframe of the project. It proved that indeed YOLOv11 is a capable model of being used in the scientific area for microplastic detection and quantification. The use of SAHI in aiding small object detection though helpful to a certain extent, improving the model's feature maps remain the key to small object detection. Image resolution is key to small object detection and generation of accurate feature maps, this barrier can be overcome through more powerful GPU chips and training with real microplastic samples. Once the YOLOv11 model is built on real urban water microplastic samples, this model can be applied alongside the use of Nile red dye to instantly detect and analyse microplastic samples on-site at the pollution area and efficiently aid pollution management.

STANDARD OPERATING PROCEDURE (SOP)

AIM: To input an image into the trained YOLOv11 model and output the image with detections.

Step 1: Open Google Colab, website <https://colab.google/>

Step 2: Click Open Colab

Step 3: Click Upload on the left side panel and drag and drop the “Microplastics detection project.py” file into the box.

Step 4: Scroll down to “Model Usage YOLOv11m-seg” heading

Step 5: In the first code cell click the play button which will appear once mouse is hovering over [] icon. Click run anyway is asked about permission setting. Wait for the log to end or a green tick beside the play button to appear, indicating cell has been ran. May take over 2 mins.

```
!pip install Ultralytics
```

Step 6: Click on the last icon, which means ‘Files’, on the left sidebar. Drag and drop the “best1280.pt” file into the “Files” section. Drag and drop your input image into the “Files” section as well and wait for them to appear.

Step 7: Click “Ok” on the Warning popup. Note that all uploaded files will be deleted after the runtime ends, which may be 30 mins if no activity is detected within the Colab website.

Step 8: Hover over the “best1280.pt” file in the “Files” section and click on the 3 dots menu icon. Click “Copy Path” and paste the path within the “ ” on this line of code. Ensure the quotation marks remain.

```
model = YOLO("best1280.pt") # .pt file of custom weights is the YOLO  
model
```

Step 9: Hover over the image file in the “Files” section and click on the 3 dots menu icon. Click “Copy Path” and paste the path within the “ ” on this line of code. Ensure the quotation marks remain.

```
results = model("/content/NM1_PXL_20250325_152417276.NIGHT.jpg",  
conf=0.1)
```

Step 10: If user wishes to see all bounding boxes regardless of confidence level, change decimal of conf=0.1 to conf=0.01 as shown below. Ensure closed bracket remains. For other confidence levels, alter the decimal value as wished.

```
results = model("/content/NM1_PXL_20250325_152417276.NIGHT.jpg",  
conf=0.01)
```

Step 10: Click the play button which will appear once mouse is hovering over [] icon. to run the cell. Play button is located next to `from ultralytics import YOLO`

Step 11: In the “Files” section, a new file named “tabular_information_output.csv” will appear. Hover and click the 3 dots for the menu icon and click Download, as once the runtime ends, files in the “Files” section will be deleted. “tabular_information_output.csv” contains a database of the coordinates of each item, the confidence level, the class ID and the corresponding class name.

Notes: *The heading of the output log file will contain the number of each class (type of plastic) present in the image.*

Output image in the log file is not saved. Hover over image, right click on mouse and select Save Image As... and save in desired location correspondingly.

AIM: To input an image into the trained YOLOv11 model and use SAHI (Slicing Aided Hyper Inference) and output the image with detections.

Step 1: Open Google Colab, website <https://colab.google/>

Step 2: Click Open Colab

Step 3: Click Upload on the left side panel and drag and drop the “Microplastics detection project.py” file into the box.

Step 4: Scroll down to “Using YOLO11m-seg + SAHI (function maths)” heading

Step 5: In the first code cell click the play button which will appear once mouse is hovering over [] icon. Click run anyway is asked about permission setting. Wait for the log to end or a green tick beside the play button to appear, indicating cell has been ran. May take over 2 mins.

```
!pip install inference inference-sdk
!pip install ultralytics
```

Step 6: Click on the last icon, which means ‘Files’, on the left sidebar. Drag and drop the “best1280.pt” file into the “Files” section. Drag and drop your input image into the “Files” section as well and wait for them to appear.

Step 7: Click “Ok” on the Warning popup. Note that all uploaded files will be deleted after the runtime ends, which may be 30 mins if no activity is detected within the Colab website.

Step 8: Hover over the “best1280.pt” file in the “Files” section and click on the 3 dots menu icon. Click “Copy Path” and paste the path within the “ ” on this line of code. Ensure the quotation marks remain.

```
model = YOLO("best1280.pt") # .pt file of custom weights is the YOLO
model
```

Step 9: Hover over the image file in the “Files” section and click on the 3 dots menu icon. Click “Copy Path” and paste the path within the “ ” on this line of code. Ensure the quotation marks remain.

```
image_file = "/content/NM1_PXL_20250325_152417276.NIGHT.jpg"
```

Step 10: Click the play button which will appear once mouse is hovering over [] icon. to run the cell. Play button is located next to `import ultralytics`

Step 11: Scroll to cell below and Click the play button which will appear once mouse is hovering over [] icon. to run the cell. Play button is located next to

```
import supervision as sv
```

Notes: *Output image in the log file is not saved. Hover over image, right click on mouse and select Save Image As... and save in desired location correspondingly.*

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